DATA-DRIVEN PIZZA SALES INSIGHTS

A COMPREHENSIVE ANALYSIS UNCOVERING TOTAL ORDERS, REVENUE TRENDS, BESTSELLING PIZZAS, CUSTOMER PREFERENCES, AND CATEGORY-WISE PERFORMANCE. FROM BASIC METRICS TO ADVANCED REVENUE INSIGHTS, THIS PROJECT REVEALS PATTERNS THAT HELP OPTIMIZE MENU STRATEGY AND BOOST BUSINESS GROWTH.

1. Project Description

This project analyzes pizza sales data to understand customer purchasing behavior, menu performance, and revenue trends. Using SQL queries across multiple relational tables, the analysis explores total orders, revenue trends, pizza category performance, customer ordering patterns, time-based demand trends, and revenue contribution by product. The goal is to uncover actionable insights that help restaurants optimize menu strategy, pricing, and operational planning.

2. Dataset Structure

- orders – Contains order ID and timestamp information.
- order_details – Contains pizza quantities ordered in each transaction.
- pizzas – Contains pizza sizes and prices.
- pizza_types – Contains pizza names and category information.
- This relational design enables analysis across product types, quantities, pricing, and revenue.

3. Business Questions Addressed

- Total number of orders placed
- Total revenue generated from pizza sales
- Identification of the highest priced pizza
- Most commonly ordered pizza size
- Top selling pizza types
- Category performance analysis
- Hourly order distribution
- Average daily order volume
- Revenue contribution by pizza type
- Cumulative revenue growth over time

4. Key Findings

- Total Orders: 21,350 orders processed.
- Total Revenue: Approximately $817,860 generated.
- Highest Priced Pizza: The Greek Pizza priced at $35.95.
- Most Popular Size: Large pizzas dominate orders.
- Top Selling Pizzas: Classic Deluxe, Barbecue Chicken, Hawaiian, Pepperoni, Thai Chicken.
- Classic pizza category generates the highest revenue share.
- Peak ordering occurs during lunch hours.
- Average daily orders are approximately 138 pizzas.

5. Business Insights

- Customers strongly prefer large pizza sizes.

- Traditional pizza varieties dominate sales.
- Lunch time represents peak demand.
- A small number of pizza types generate most revenue.

6. Business Recommendations

- Highlight best selling pizzas in menu design.
- Promote high revenue items such as Thai Chicken and Barbecue Chicken pizzas.
- Introduce lunch promotions to increase order value.
- Offer bundle deals and family packages for large pizzas.
- Review and optimize low performing menu items.